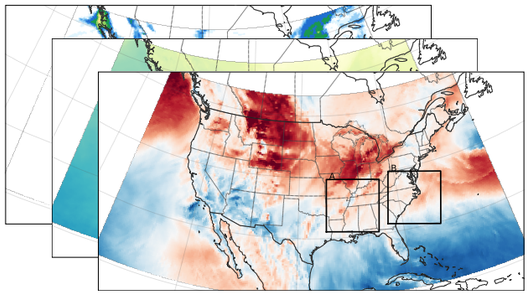
MERRA-Flow: GENERATIVE CONUS DOWNSCALING

GEOS-FP (~25 km) → Conditional flow matching → GEOS grid (~3 km)

A COARSE INPUTS

GEOS-FP · ~25 km



Weather at t − 1 h and t
Rain rate · PRECTOT
10 m winds · U10M, V10M
Column vapor · TQV
2 m humidity · QV2M
Sea-level pressure · SLP

Coarse baselines at t
T2M, PRECTOT, U10M, V10M
Surface pressure · PS

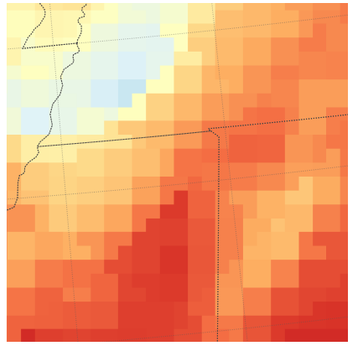
Static fields
Elevation · lat/lon · cell area
Land/lake/ocean fractions
Water distance · x/y slopes
Lake-availability mask

Time (sine/cosine)
Annual · UTC · local solar

B RESOLUTION

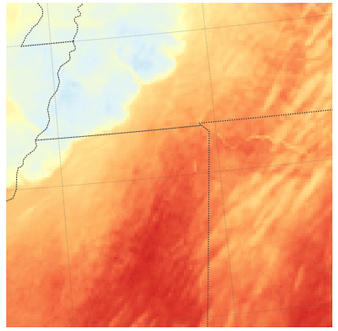
Same extent · same scale

GEOS-FP · ~25 km





GEOS replay · ~3 km

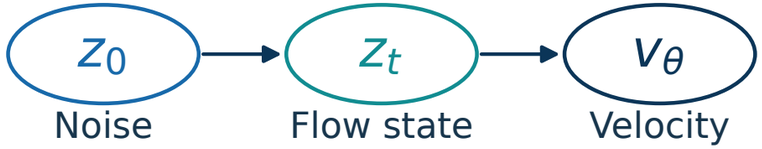




Temperature (°C) · 3× zoom

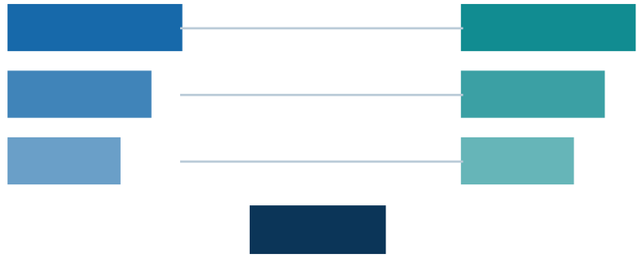
C FLOW MATCHING + ENSEMBLE GENERATION

Frozen regression · local patches + broad context



$$z_t = (1 - t)z_0 + tz_1$$

Conditional U-Net

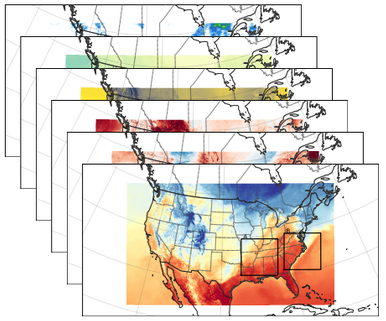


Weighted velocity loss



Core + halo
Area + channels

Six output fields



~3 km / member

T_2m · rain · p_s
u_{10} · v_{10} · q_2m

8 members
Independent seeds

Tiled Heun → decode → ensemble

Blend tile velocities; advance one shared domain state